

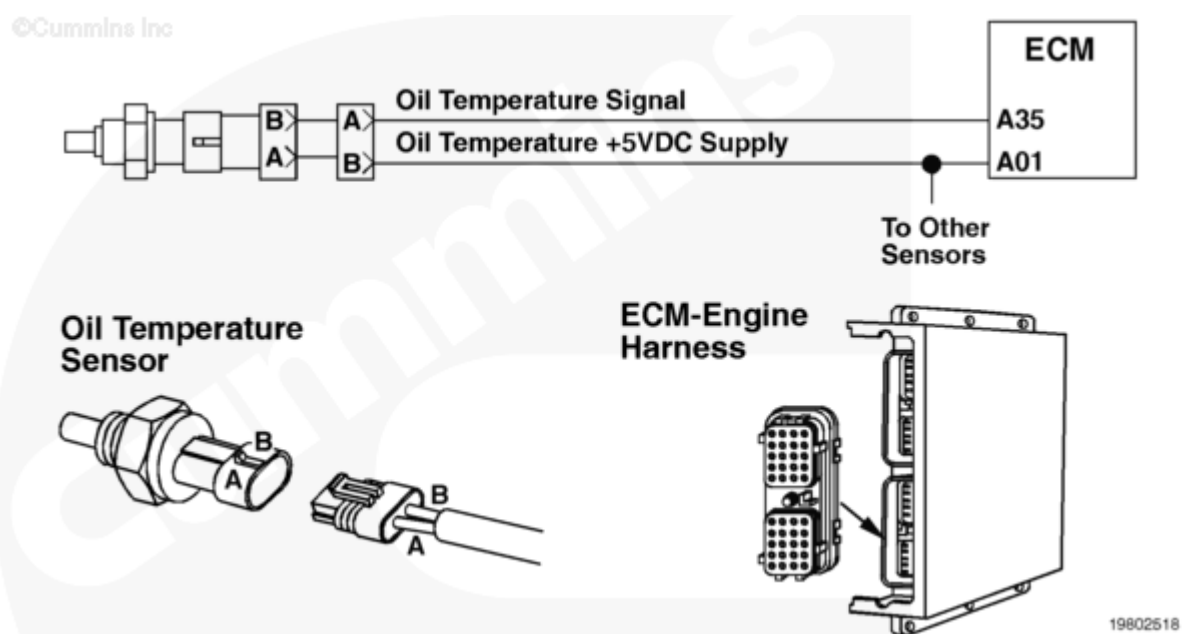
Fault Code: 213

Engine Oil Temperature Sensor 1 Circuit - Voltage Below Normal or Shorted to Low Source

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Overview

Codes	Reason	Effect
Fault Code: 213 PID(P): 175 SPN: 175 FMI: 4 Lamp: Yellow SRT: 00-502	Engine Oil Temperature Sensor 1 Circuit - Voltage Below Normal or Shorted to Low Source. Low voltage detected on SIGNAL pin 35 of the main harness A ECM connector.	Fault Code 214 is disabled.



Oil Temperature Sensor Circuit

Circuit Description

The oil temperature sensor circuit provides the oil temperature signal to the ECM. The resistance of the oil temperature sensor varies with temperature. The ECM detects the change in resistance of the oil temperature sensor by monitoring the voltage across an internal resistor that is in series with the oil temperature sensor. The change in voltage across the internal resistor is translated into a temperature change.

Component Location

The oil temperature sensor is located on the right side of the oil pan.

Shoptalk

- The resistance of the sensor varies with the temperature. The reading that you observe should compare to the following table if the sensor is functioning properly.

Temperature	Temperature	Resistance
(°C)	(°F)	(ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-213 (</qs3/pubsys2/xml/en/procedures/81/81-t05-213.html>)

Last Modified: 07-Jul-2015
